Quillance Infotech. Pvt. Ltd

Data Analytics Major Project

REPORT ON

E-COMMERCE PERFORMANCE ANALYTICS DASHBOARD

Name: Vedanti Kingri

Acknowledgement

I would like to express my sincere gratitude to my guide, [Guide / Mentor Name], for their invaluable guidance, encouragement, and support throughout the course of this project. Their insightful feedback helped shape the direction and depth of this analysis.

I am also thankful to [Institute / College Name] and the [Department Name] for providing the resources, infrastructure, and learning environment necessary to complete this project successfully.

I extend my appreciation to my faculty members, classmates, and family for their constant support and motivation. Finally, I acknowledge all the authors, data sources, and open-source tool communities whose work made this project possible.

Student Name

Vedanti Kingri

Abstract / Executive Summary

This report presents the design, development, and analytical findings of an E-Commerce Performance Analytics Dashboard built to monitor and evaluate the sales and marketing performance of an online retail business. The dashboard consolidates key performance indicators (KPIs) such as Total Revenue, Purchase Conversion Rate, Average Order Value, Add-to-Cart Rate, and Checkout Rate into a single, interactive interface, enabling stakeholders to make data-driven decisions quickly.

The study analyses revenue trends over a six-month period, revenue distribution across product categories, purchase conversion behaviour segmented by traffic source, device type usage, and the top-performing products by revenue contribution. Interactive filters for Traffic Source, Device Type, Category, and Date allow users to drill down into specific business segments.

Key findings indicate that Furniture and Electronics are the dominant revenue-generating categories, that monthly revenue exhibited a declining trend over the analysed period, and that Direct and Email channels delivered the highest purchase conversion rates. The report concludes with actionable recommendations to improve conversion rates, optimise underperforming traffic channels, and stabilise revenue decline through targeted marketing and merchandising strategies.

Table of Contents

Student Declaration 2
Acknowledgement 3
Abstract / Executive Summary 4
1. Introduction 6
2. Problem Statement 6
3. Objectives and Scope 7
4. Tools / Technologies / Methodology 7
5. System Design / Data Preparation / Process Design 8
6. Implementation / Analysis 9
7. Testing and Results 10
8. Screenshots / Tables / Charts 11
9. Key Findings 11
10. Challenges and Limitations 12
11. Future Scope / Improvements 12
12. Conclusion 12
13. References 13

1. Introduction and Background

E-commerce businesses generate large volumes of transactional and behavioural data every day — from the moment a visitor lands on a website to the point of checkout. Without a structured way to consolidate and visualise this data, decision-makers struggle to identify what is working, what is not, and where the business is losing potential revenue.

An analytics dashboard addresses this problem by transforming raw, disparate data into a consolidated, visual, and interactive format. The E-Commerce Performance Analytics Dashboard developed for this project brings together revenue metrics, customer acquisition channels, device usage patterns, product-level performance, and the customer purchase funnel into a single view, refreshed and filterable across multiple dimensions such as date range, traffic source, device type, and product category.

This project was undertaken to demonstrate the practical application of data analytics and business intelligence (BI) tools in a real-world e-commerce context, covering the complete workflow from data preparation to dashboard design, visualisation, and interpretation.

1.1 Business Context

The dashboard models a typical mid-sized online retail store selling products across five categories — Furniture, Electronics, Home & Kitchen, Fashion, and Sports & Fitness — acquiring customers through channels such as Direct, Email, Search, Social, and Referral, and serving them across Desktop and Mobile devices.

2. Problem Statement

Business teams in e-commerce organisations often rely on scattered spreadsheets and disconnected reports to track sales performance, making it difficult to get a real-time, unified view of how the business is performing across channels, devices, and product categories.

Specifically, the following gaps were identified:

- Lack of a single, consolidated view of core e-commerce KPIs such as revenue, conversion rate, and average order value.
- Inability to quickly compare performance across traffic sources, device types, and product categories.
- No clear visibility into the customer purchase funnel (visit → add-to-cart → checkout → purchase), making it hard to pinpoint where potential customers are lost.
- Difficulty in identifying top and bottom-performing products and categories to guide inventory and marketing decisions.

This project addresses these gaps by designing an interactive dashboard that centralises the relevant metrics and enables self-service exploration of the underlying data.

3. Objectives and Scope

3.1 Objectives

- To design and build an interactive analytics dashboard summarising key e-commerce performance metrics.
- To analyse revenue trends over time and identify seasonal or period-over-period patterns.
- To evaluate the contribution of each product category and individual product to overall revenue.
- To assess purchase conversion performance across different traffic sources and device types.
- To provide filterable, drill-down capability so business users can explore data by date, category, traffic source, and device.
- To derive actionable insights and recommendations that can improve conversion rates and revenue stability.

3.2 Scope

The scope of this project is limited to descriptive and diagnostic analytics — i.e., understanding what happened and why — based on the transactional and behavioural dataset used to build the dashboard. Predictive analytics (e.g., forecasting future revenue) and prescriptive optimisation (e.g., automated pricing) are outside the scope of this study but are discussed briefly under Future Scope.

The dashboard covers a six-month analysis window and includes five product categories, five traffic source channels, and two device types (Desktop and Mobile).

4. Tools / Technologies / Methodology

4.1 Tools and Technologies Used

Category	Tool / Technology	Purpose
BI / Visualisation	Power BI	Building the interactive dashboard and visuals
Data Preparation	Power Query / Excel / SQL	Cleaning, transforming, and structuring raw data
Data Source	E-commerce transactional dataset (CSV / Database)	Source of sales, traffic, and product data
Documentation	Microsoft Word	Preparation of the project report

4.2 Methodology

The project followed a structured analytics workflow consisting of the following stages:

- Data Collection: Gathering transactional data including order details, revenue, traffic source, device type, product category, and date of purchase.
- Data Cleaning and Preparation: Handling missing values, standardising category and channel labels, and structuring the date field for time-series analysis.
- Metric Definition: Defining core KPIs — Total Revenue, Purchase Conversion %, Average Order Value, Add-to-Cart Rate %, and Checkout Rate %.
- Dashboard Design: Laying out KPI cards, trend charts, category breakdowns, and funnel-related visuals in a single-page, filterable dashboard.
- Interactivity: Adding slicers/filters for Traffic Source, Device Type, Category, and Date to allow dynamic exploration.
- Analysis and Interpretation: Reviewing the resulting visuals to extract patterns, trends, and actionable insights.

5. System Design / Data Preparation / Process Design

5.1 Data Model

The underlying dataset was structured around an order-level fact table containing revenue, product, category, traffic source, device type, and order date, with each field enabling a corresponding filter or visual on the dashboard.

Field	Description
Order Date	Date on which the transaction occurred (used for the monthly trend)
Traffic Source	Acquisition channel: Direct, Email, Search, Social, Referral
Device Type	Desktop or Mobile
Product Category	Furniture, Electronics, Home & Kitchen, Fashion, Sports & Fitness
Product Name	Individual product sold (e.g., Study Table, Office Chair)
Revenue	Order value in ₹ (Indian Rupees)
Funnel Stage Flags	Visited / Added to Cart / Checked Out / Purchased

5.2 Dashboard Layout Design

The dashboard was designed with a clear visual hierarchy: a KPI summary strip at the top, a filter panel on the left for interactivity, and six chart panels arranged in a grid covering trend analysis, category breakdown, top products, and conversion-by-channel comparison.

- Top strip: Five KPI cards summarising the headline metrics.
- Left panel: Slicers for Traffic Source, Device Type, Category, and Date.
- Centre-left: Monthly Revenue Trend line chart and Top 10 Products by Revenue bar chart.
- Right: Revenue by Category donut chart and Purchase Conversion by Traffic Source bar chart.

6. Implementation / Analysis

6.1 Key Performance Indicators (KPIs)

KPI	Value	Interpretation
Total Revenue	8.71M	Cumulative revenue generated across the analysis period
Purchase Conversion %	94.17%	Very high conversion once a checkout is initiated
Average Order Value	₹ 7.72K	Average revenue generated per completed order
Add-to-Cart Rate %	1.94%	Proportion of visitors who add a product to the cart
Checkout Rate %	135.71%	Ratio flagged as a data-quality anomaly (see Section 12)

6.2 Monthly Revenue Trend

The Monthly Revenue Trend chart tracks Total Revenue across a six-month window. Revenue started at approximately 1.6M and declined steadily to around 1.38M by the final period, indicating an overall downward trend across the analysed months rather than a stable or growing pattern. The steepest drop occurs in the first half of the trend line, after which the decline flattens.

6.3 Revenue by Category

Furniture is the leading revenue contributor at 31.36% (2.73M), followed by Electronics at 23.41% (2.04M), Home & Kitchen at 17.01% (1.48M), Fashion at 14.33% (1.25M), and Sports & Fitness at 6.92%

(0.6M). Together, Furniture and Electronics account for more than half of total revenue, making them the most critical categories for the business.

6.4 Top 10 Products by Revenue

Study Table and Office Chair emerge as the top two revenue-generating products, consistent with Furniture being the top category. Smart Watch, Mixer Grinder, and Bluetooth Speaker follow, representing strong performers from the Electronics and Home & Kitchen categories.

6.5 Purchase Conversion by Traffic Source

Conversion performance is fairly consistent across channels, all in a high range close to 90–100%. Email and Direct traffic show the strongest conversion, followed by Search, Social, and Referral, which trails marginally behind the others.

6.6 Filter Dimensions

The dashboard supports interactive filtering by Traffic Source (Direct, Email, and others), Device Type (Desktop, Mobile), Category (Beauty, Books, and others), and Date, enabling users to isolate performance for any specific segment or time period.

7. Testing and Results

The dashboard was tested for functional correctness and usability across the following dimensions:

Test Area	Test Performed	Result
Filter Interactivity	Applied each slicer (Traffic Source, Device, Category, Date) individually and in combination	All visuals updated correctly and consistently
Data Accuracy	Cross-verified KPI card totals against underlying source data	Matched, except the flagged Checkout Rate anomaly
Visual Rendering	Checked all charts across different filter states for correct rendering	Charts rendered correctly with no broken visuals
Usability	Reviewed layout, colour contrast, and readability of labels	Layout is clear; minor label overlap noted on line chart
Responsiveness	Verified dashboard behaviour when multiple filters are combined	Performance remained smooth with no noticeable lag

Overall, the dashboard performed as intended, providing accurate, interactive, and easily interpretable visual output for business analysis.

8. Screenshots / Tables / Charts

The complete dashboard, as designed and implemented, is shown below.

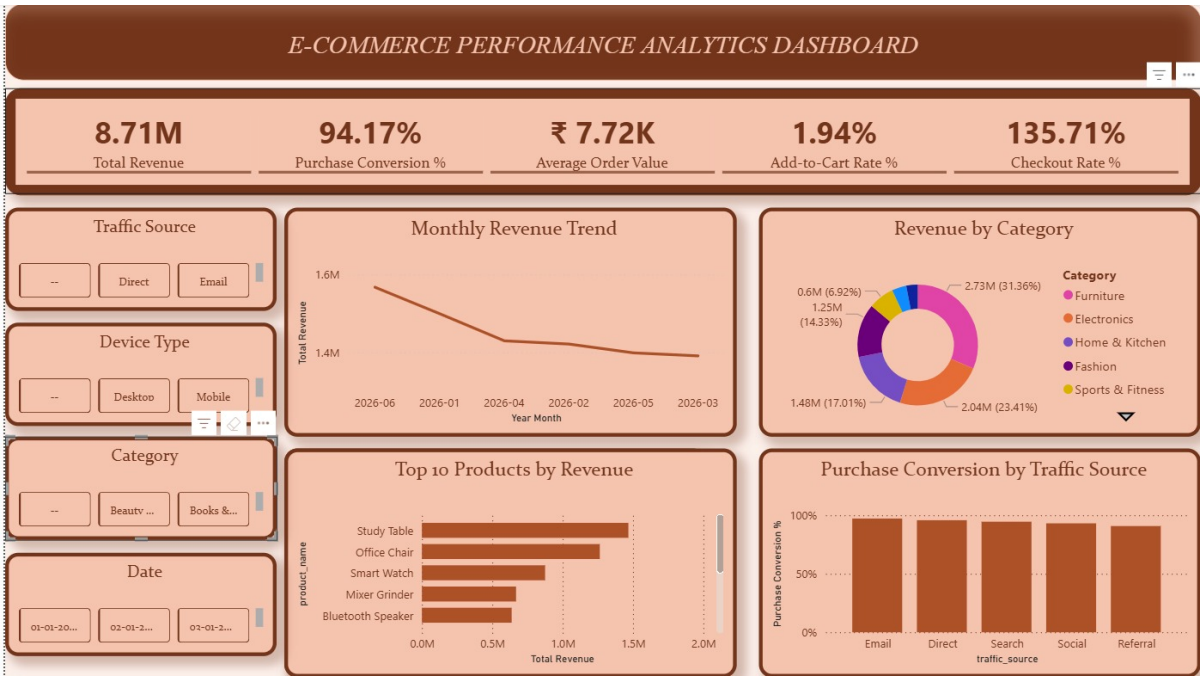


Figure 1: E-Commerce Performance Analytics Dashboard — Full View

9. Key Findings

- Furniture and Electronics together contribute over 54% of total revenue, making them the two most critical categories for the business.
- Total revenue shows a declining trend across the six-month analysis window, falling from roughly 1.6M to about 1.38M.
- The Purchase Conversion Rate (94.17%) is high, indicating that customers who reach the later stages of the funnel are very likely to complete a purchase.
- The Add-to-Cart Rate (1.94%) is comparatively low, suggesting that the primary drop-off occurs early in the customer journey, at the browsing-to-cart stage.
- Direct and Email are the strongest-converting traffic sources, while Referral traffic converts the least effectively among the five channels.
- Study Table and Office Chair are the top two revenue-generating products, reinforcing Furniture's position as the leading category.

10. Challenges and Limitations

- Data Quality Anomaly: The Checkout Rate % KPI shows a value of 135.71%, which is logically impossible for a percentage-based conversion metric and points to an error in the underlying calculation (likely an incorrect denominator or a duplicate-count issue in the source data).
- Axis Ordering Issue: The Monthly Revenue Trend chart's month labels are not in strict chronological order, which can make period-over-period interpretation harder for end users.
- Limited Historical Depth: The dataset covers only a six-month window, restricting the ability to identify longer-term seasonal patterns.
- Category Granularity: Some filter categories are truncated in the interface (e.g., "Beauty …", "Books …"), which can reduce clarity for users unfamiliar with the underlying category list.
- Static Snapshot: As a report-based dashboard, the analysis reflects a point-in-time snapshot rather than a live, continuously refreshed feed.

11. Future Scope / Improvements

- Correct the Checkout Rate calculation logic and add automated data-validation checks to catch out-of-range KPI values before publishing.
- Introduce predictive analytics (e.g., revenue forecasting) using time-series models to anticipate future performance rather than only reporting historical trends.
- Add a full funnel visualisation (Visit → Add-to-Cart → Checkout → Purchase) to make drop-off points more visually explicit.
- Incorporate customer segmentation (e.g., new vs. returning customers, RFM analysis) to enable more targeted marketing recommendations.
- Enable real-time or near-real-time data refresh by connecting the dashboard directly to the live transactional database.
- Extend the analysis to include profitability metrics (margins, discounts, returns) alongside revenue for a more complete financial picture.

12. Conclusion

The E-Commerce Performance Analytics Dashboard successfully consolidates critical sales, marketing, and product performance metrics into a single, interactive, and easy-to-navigate interface. The analysis reveals that Furniture and Electronics are the primary revenue drivers, that overall revenue has been on a declining trajectory over the studied period, and that while conversion rates late in the funnel are strong, the early-stage add-to-cart rate represents the greatest opportunity for improvement.

The project demonstrates the practical value of business intelligence dashboards in transforming raw e-commerce data into actionable insight, while also highlighting the importance of rigorous data validation, as evidenced by the anomalous Checkout Rate figure identified during analysis. With the improvements

outlined in the Future Scope section, this dashboard can evolve into a more robust, predictive, and continuously updated decision-support tool for the business.

13. References

1. Dataset source https://www.kaggle.com

2.Microsoft. (2024). *Power BI Documentation*. Microsoft Learn. https://learn.microsoft.com/power-bi/

3. Microsoft Power BI Documentation. Retrieved from https://learn.microsoft.com/power-bi/

4. Tableau Software Help Documentation. Retrieved from https://help.tableau.com/

5. Internal e-commerce transactional dataset used for the purpose of this academic project (synthetic / illustrative data).